



Figure 1: Motion UI

Browser based IDEs

Traditionally, Web development IDEs have been used with desktop applications. For the past few years, we have seen numerous browser based cloud IDEs being developed rapidly. These IDEs don't require any software other than a Web browser, which allows developers to write code from any computer, over the Internet.

Some popular browser based IDEs are Cloud9, Aptana, Mozilla's WebIDE, Codiad, Codebox, Orionhub and Icedoer. In this list, Codiad, Codebox and Orionhub are 100 per cent open source. There are restrictions in the Cloud9 free version; we can use only one private workspace.

Elasticsearch (Elastic), a distributed search engine

Elasticsearch is a highly scalable open source full-text search and analytics engine. Elasticsearch allows you to store, search and analyse large volumes of data quickly and in near real-time. It provides a distributed, multi-tenant-capable full-text search engine with a RESTful Web interface and schema-free JSON documents. Elasticsearch is developed in Java, and is released as open source under the terms of the Apache License. Developed by Shay Banon, the first version of Elasticsearch was released in February 2010.

Some of the well-known users of Elasticsearch are Wikimedia, GitHub, Stack Exchange (StackOverflow), StumbleUpon, Mozilla, Quora, Madeus IT Group, Foursquare, Etsy, SoundCloud, FDA, CERN, Centre for Open Science, Reverb, Netflix and Pixabay.

Docker

Docker is a new container technology. It's very popular nowadays because it allows users to run multiple apps on the same old servers, and it also makes it very easy to package and ship programs. A Docker container has all the dependencies it requires to run an application.

Docker automates the deployment of applications inside software containers, by providing an additional layer of abstraction and automation of operating-system-level virtualisation on Linux. Docker uses the resource isolation features of the Linux kernel, such as cgroups and kernel namespaces, and a union-capable file system such as aufs, as

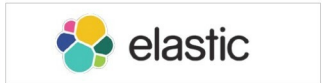


Figure 2: Elasticsearch

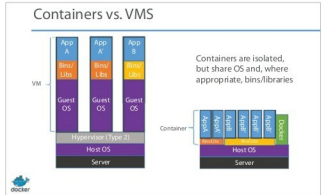


Figure 3: Containers vs VMs

well as other features. This allows independent 'containers' to run within a single Linux instance, avoiding the overhead of starting and maintaining virtual machines.

Responsive design

Responsive Web design is becoming most important as mobile traffic is increasing day by day. This is also a must-have feature in your website, because Google is boosting the rankings of mobile-friendly pages on mobile search results. So if your site's pages aren't mobile-friendly, there may be a significant decrease in mobile traffic from Google Search. Responsive design helps Google to provide the relevant, high quality search results that are optimised for specific devices.

Responsive design is also no longer restricted to being just 'mobile-friendly' or to mobile responsiveness alone. There are already plenty of other devices, screens and integrated wearable gadgets like the Apple watch, Google Glass and Oculus Rift which are much talked about technologies.

There are some popular open source frameworks which have developed responsive design like Bootstrap, Foundation, Skeleton, YAML 4, Tuktuk, Less Framework, etc.



Figure 4: Responsive design